SnapBudget

Snap. Extract. Save. Zero typing required.

Problem Statement ID: FM-01

Team: CodeX

Theme: FinTech

The Challenge & it's *Solution*

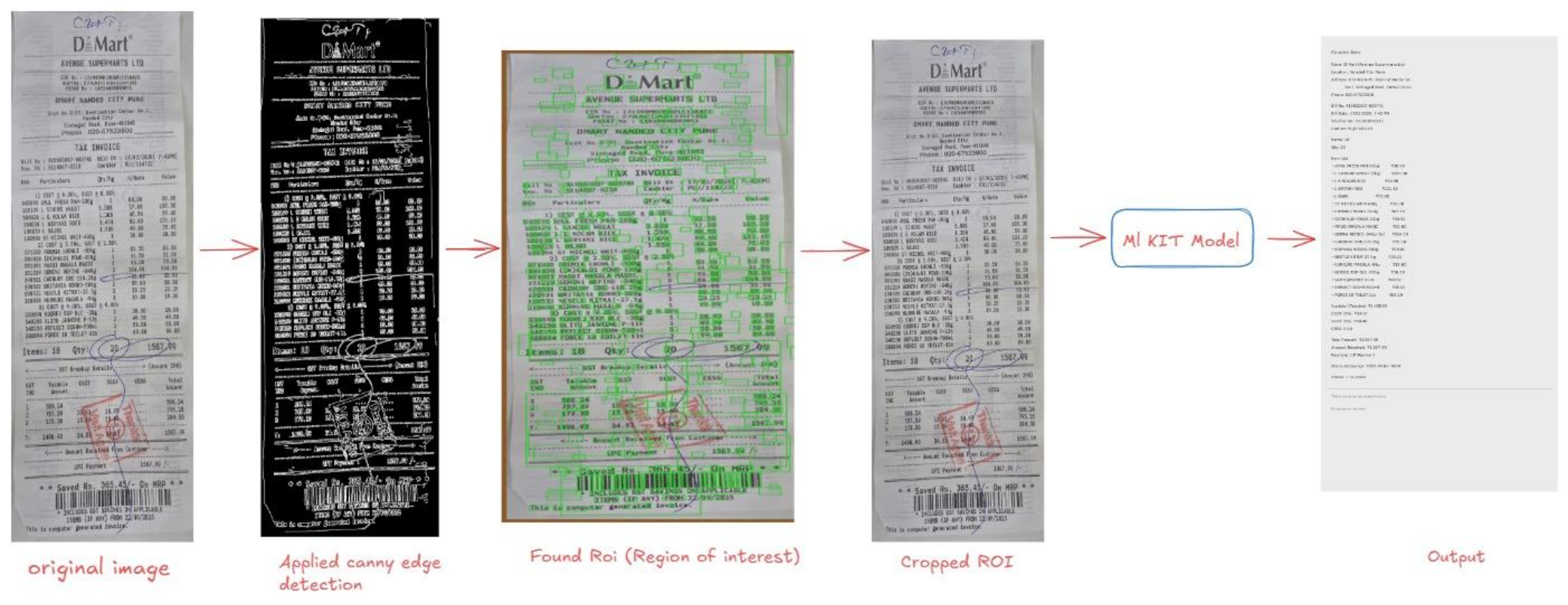
Problem vs SnapBudget Solution

Why existing tools fail — and how we fix it

THE PROBLEM		SNAPBUDGET SOLUTION
Manual entry takes 10+ minutes per receipt People give up within a week		One tap Snap button Full extraction in under 2 seconds
Single-pass OCR fails on blurry or noisy Indian receipts		7-stage hybrid pipeline with Llama 4 vision fallback for blurry images
Raw text scan misreads amounts B vs 8, O vs 0 character errors	VS	Dedicated OCR Error Correction layer fixes character confusion automatically
No Indian GST support No GSTIN validation or Rs parsing		Full GSTIN extraction + CGST/SGST split Built for Indian retail chains
Scanning same receipt twice creates duplicate expense entries		Fuzzy Duplicate Detector using Levenshtein distance scoring
Cloud-only apps send all data to servers Privacy and cost concerns		90% offline — data stays on device Cloud AI only when user opts in

Technical Execution: Digitization Pipeline

The receipt digitization process follows six sequential steps, from initial image capture through machine learning text recognition to structured data output.



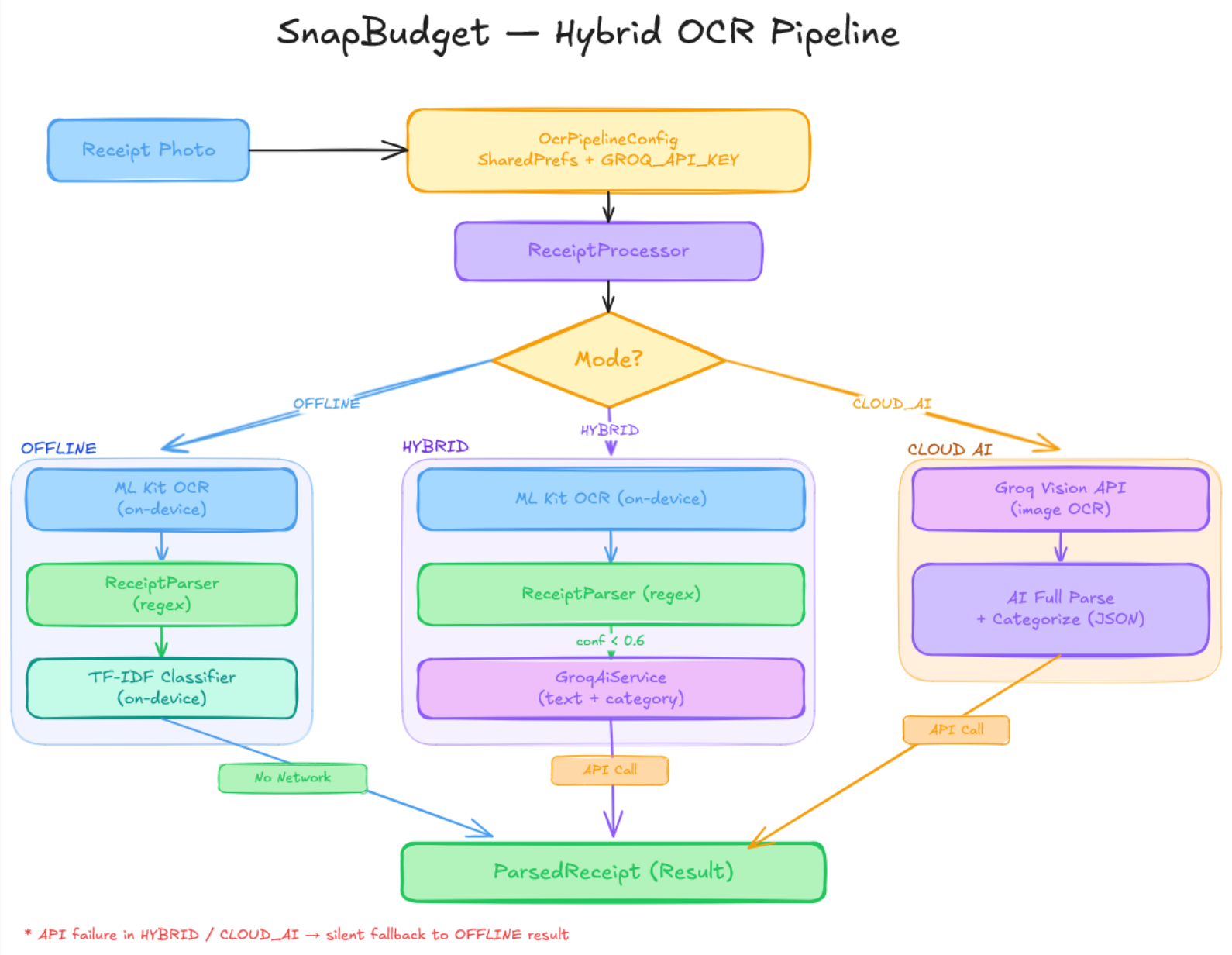
Hybrid AI Architecture

ON-DEVICE

- Privacy-first local processing in <2s
- Receipt photo, ML Kit OCR v2, Spatial Parser

CLOUD AI

- Groq API, Llama 3.3 & 4 Vision for complex cases
- User-authorized only, 95%+ accuracy



Comprehensive Technology Stack

FRONTEND

UI / Frontend

Kotlin, Jetpack Navigation, Material Design 3

LOCAL AI

Local Intelligence

ML Kit Text Recognition v2, TF-IDF Classifier, Regex Parser

CLOUD AI

Cloud AI Brain

Groq Vision API, Llama Llama 3.3 (70B), Llama 4 Scout

DATA

Data & Core

Room Database, Kotlin Coroutines & Flow, Encrypted SharedPrefs

User Impact & Key Metrics

PROCESSING SPEED

< 2s

Time to process receipts and transactions

ACCURACY

> 95

Data extraction and categorization accuracy

OFFLINE CAPABILITY

90%

Tasks performed without internet connection

TIME SAVED

15-20 min

Weekly time savings per user

SERVER COST

Rs 0

Per user due to local-first architecture

TARGET USERS

4 Segments

Employees, Business Owners, Students, Families

KEY BENEFITS

Benefit Breakdown

ECONOMIC

Free Tier + GST Auto-Extract

EFFICIENCY

Instant AI-powered expense tracking and categorization

SECURITY

Local-only processing ensures financial data never leaves your device

PRIVACY

Offline-First, No Cloud Storage

ACCESSIBILITY

Indian GST, Works Offline

ACCURACY

Advanced hybrid OCR for precise merchant, date, and amount detection

SOCIAL

Financial Discipline for 100M+ Users

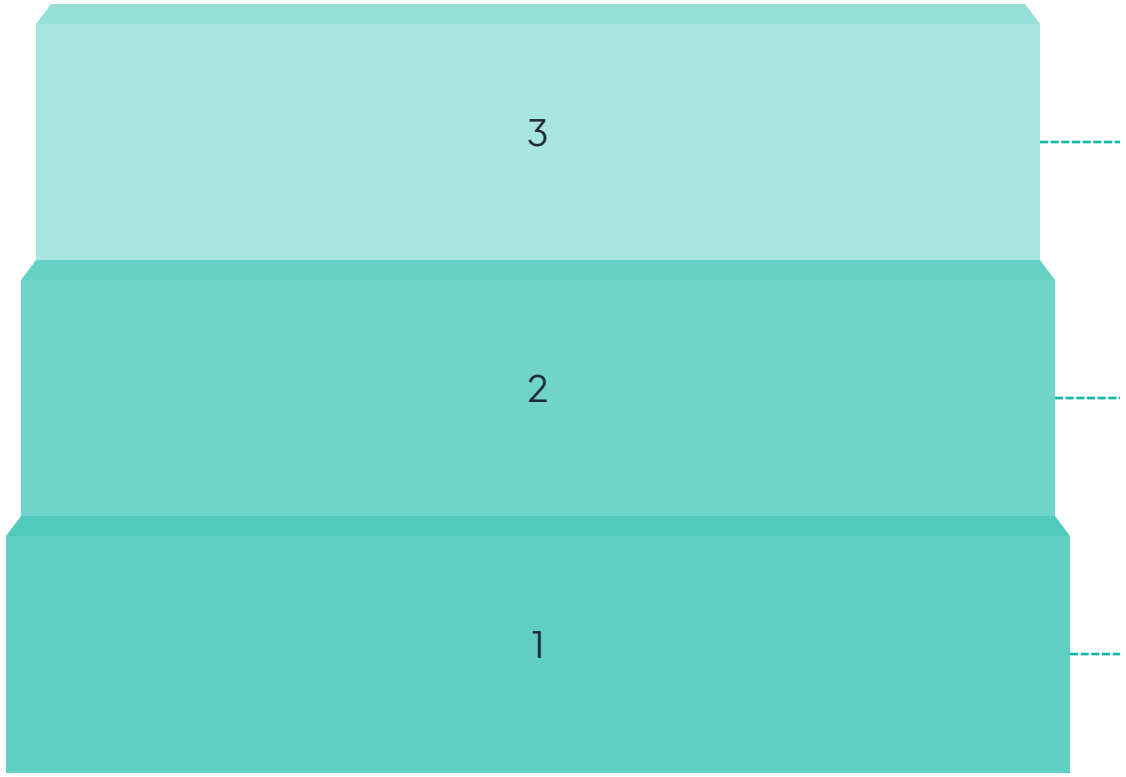
CONVENIENCE

Fast 1-tap Google Sign-In with a personalized user dashboard

SIMPLICITY

Frictionless, one-tap receipt capture via the centralized scan button

Product Roadmap: From 10-Minute Chore to 2-Second Habit



Phase 3: Bank SMS cross-referencing, UPI payment integration, multi-user family accounts, vernacular support

Phase 2: Subscription auto-detection, split-bill feature, CSV/PDF export, AI category auto-tagging

Phase 1: Hybrid OCR, GSTIN extraction, fuzzy duplicate detection, working prototype

SnapBudget transforms expense tracking from a tedious 10-minute chore into a seamless 2-second habit